# HeatGeo: Corpus-Level Heat-Kernel Geometry Distillation for Compact Text Embedding Models

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## Abstract

Large text embedding models provide strong semantic representations, but their inference cost limits their use in retrieval, clustering, and retrieval-augmented generation systems that must encode large or frequently changing corpora. Knowledge distillation is a natural way to compress such models, yet existing embedding distillation objectives mainly transfer pointwise teacher embeddings, mini-batch similarity matrices, token-level representations, or mixtures of local losses. These signals are useful, but they do not explicitly supervise the corpus-level neighborhood geometry on which embedding models are ultimately evaluated. We propose **HeatGeo**, a corpus-level geometry distillation framework that transfers the teacher's semantic diffusion process to a compact student encoder. HeatGeo first builds a mutual $k$-nearest-neighbor graph over teacher embeddings for the training corpus, then derives multi-scale heat-kernel diffusion distributions through random walks on this graph. The student is trained to match these diffusion distributions over sampled candidate neighborhoods, while a spectral manifold objective anchors global topology using graph Laplacian eigenvectors. HeatGeo requires only cached teacher embeddings, introduces no architectural changes to the student, and is naturally compatible with heterogeneous teacher and student tokenizers. We evaluate HeatGeo on semantic textual similarity, text classification, and pair classification benchmarks, testing whether corpus-level diffusion supervision better preserves retrieval-oriented geometry than pointwise, token-level, and mini-batch relational distillation baselines.


## Introduction

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. In these systems, a model is not judged only by whether it embeds one sentence close to a teacher vector; it is judged by whether the resulting corpus index organizes examples into useful neighborhoods. Strong recent embedding models, including LLM-based encoders, improve this organization but are costly to run at scale. Encoding large private corpora, refreshing indexes as documents change, or serving retrieval on constrained hardware therefore motivates compact student encoders that inherit the semantic structure of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, and relational losses that match pairwise similarities within a mini-batch. These objectives are attractive because they are simple and efficient. They also fit naturally into stochastic training.

However, embedding models are deployed in a different regime from the one these losses directly supervise. Retrieval and clustering depend on the neighborhood structure induced over an entire corpus. A pointwise loss asks the student to reproduce teacher coordinates, even when a compact student may not have enough capacity to match the teacher's absolute geometry. A mini-batch relational loss improves on this by comparing local similarity patterns, but it sees only the examples that happen to co-occur in a batch. Many relations that matter for retrieval are neither pointwise nor confined to a single batch: samples may be connected through multi-hop semantic paths, and useful clusters often emerge from the accumulated geometry of the full corpus.

This mismatch motivates the central question of this work: can we distill a large embedding teacher by matching its corpus-level semantic diffusion process, rather than only matching individual vectors or mini-batch relations? The intuition is simple. If the teacher's embedding space defines a semantic manifold over the training corpus, then random walks on a teacher-induced neighborhood graph reveal how semantic mass flows from each example to nearby examples, broader local regions, and higher-order communities. Matching this diffusion process gives the student direct supervision on the retrieval geometry it must later reproduce.

We propose **HeatGeo**, a corpus-level heat-kernel geometry distillation method for compact text embedding models. HeatGeo first caches teacher embeddings for the training corpus and constructs a mutual $k$-nearest-neighbor graph using teacher cosine similarity. From this graph, it defines a transition matrix and precomputes multi-scale diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

HeatGeo adds two stabilizing signals. First, a spectral manifold objective projects student embeddings into low-dimensional spectral coordinates computed from the teacher graph Laplacian, encouraging the student to preserve global semantic topology rather than only local neighbors. Second, a lightweight pointwise anchor aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. A simple curriculum introduces diffusion scales gradually: the student first learns local neighborhoods, then semantic regions, and finally longer-range diffusion with spectral anchoring.

Our contributions are:

- We formulate text embedding distillation as a corpus-level geometry transfer problem, highlighting the gap between common pointwise or mini-batch losses and corpus-level retrieval behavior.
- We introduce HeatGeo, which distills multi-scale heat diffusion distributions from a teacher-induced mutual $k$NN graph into a compact student encoder.
- We propose a spectral manifold anchoring objective that transfers global topology from the teacher graph without requiring teacher hidden states or modifying the student architecture.
- We provide an evaluation protocol for HeatGeo across semantic textual similarity, text classification, and pair classification benchmarks, with ablations for diffusion scale, spectral anchoring, candidate sampling, and curriculum.

## Related Work

**Text Embedding Models and Evaluation.** Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers and Gurevych 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao, Yao, and Chen 2021). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al. 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al. 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al. 2025), and Dewey targets long-context embedding for RAG-style retrieval (Zhang, Zou, and Zhou 2025). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should preserve the teacher's corpus-level neighborhood organization. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills the semantic graph induced over the training corpus.

**Distillation for Language and Embedding Models.** Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al. 2024; Ko et al. 2025; Anshumann et al. 2025; Kim et al. 2025; Xu et al. 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al. 2024b; Chen et al. 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al. 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al. 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic and Rueckstiess 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al. 2026). These methods establish distillation as a central recipe for embedding model development. Their supervision, however, is still primarily defined by individual teacher vectors, generated/relabelled pairs, task labels, dimensional heads, or local teacher scores. HeatGeo changes the distillation object from per-example or per-pair behavior to multi-scale diffusion over a teacher-induced corpus graph.

**Local and Relational Supervision.** Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher's absolute coordinates even when the student's capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor to nearby regions of the full corpus. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al. 2019; Alqahtani et al. 2021; Dao et al. 2026). Yet they still optimize local representational agreement rather than a corpus-level retrieval geometry. HeatGeo keeps the useful architecture-agnostic property of final-embedding distillation, but replaces mini-batch relational snapshots with cached diffusion targets sampled from the full teacher graph.

**Graph, Manifold, and Diffusion Perspectives.** Graph-based manifold learning studies how neighborhood graphs reveal lower-dimensional structure in high-dimensional data. Laplacian Eigenmaps use graph Laplacian eigenvectors to preserve local neighborhoods in low-dimensional representations (Belkin and Niyogi 2003), and Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman and Lafon 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al. 2025; Zhang et al. 2022). These works show that graph structure and diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph and the spectral coordinates are not used as a standalone dimensionality-reduction output. Instead, the graph is induced by a strong text embedding teacher over the training corpus, diffusion distributions become soft neighborhood targets for the student, and Laplacian eigenvectors act as a global topology anchor. This directly addresses the gap left by pointwise, token-level, layer-level, and mini-batch relational distillation: compact text encoders are trained to preserve the teacher's corpus-level semantic diffusion process, the object most closely aligned with retrieval and clustering behavior.

## Problem Setup

Let $\mathcal{D} = \{x_i\}_{i=1}^{N}$ denote a training corpus. A teacher encoder $T$ maps each input to a teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \tag{1}$$

and a compact student encoder $S_\theta$ maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \tag{2}$$

The goal is to train $S_\theta$ so that the student embedding space preserves the semantic organization induced by the teacher, especially nearest-neighbor rankings, local semantic regions, and higher-order corpus structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \cos(Ws_i, t_i)\right), \tag{3}$$

where $W$ is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch $B$:

$$\mathcal{L}_{\text{rel}} = \left\| G_B^S - G_B^T \right\|_F^2, \tag{4}$$

where $G_B^S$ and $G_B^T$ are cosine-similarity matrices over examples in $B$. HeatGeo generalizes the relational view from mini-batch snapshots to a precomputed corpus-level teacher graph.

## HeatGeo

### Teacher Semantic Graph

HeatGeo first encodes the full training corpus with the teacher and constructs a teacher semantic graph. For each example $x_i$, let

$$\mathcal{N}_k^T(i) = \text{TopK}_{j\neq i}\cos(t_i, t_j) \tag{5}$$

be its top-$k$ teacher nearest neighbors. To reduce noisy one-directional edges and hubness, HeatGeo uses a mutual $k$NN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \tag{6}$$

The weighted adjacency matrix $W^T$ is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

where $\tau_g$ controls graph sharpness. Let $D^T$ be the diagonal degree matrix. The row-normalized transition matrix is

$$P^T = (D^T)^{-1}W^T. \tag{8}$$

Each row $P_i^T$ defines a one-step semantic random-walk distribution from anchor $x_i$.

### Multi-Scale Heat Diffusion Targets

A one-step transition captures direct neighbors, but embedding spaces also contain higher-order semantic structure. HeatGeo therefore uses multiple diffusion scales:

$$P_r^T = (P^T)^r, \qquad r \in \mathcal{R}, \tag{9}$$

where $\mathcal{R}$ is typically $\{1, 2, 4\}$. Small $r$ preserves sharp local neighborhoods, while larger $r$ captures semantic regions and multi-hop communities. In implementation, diffusion is approximated sparsely by propagating probability mass through graph neighbor lists and retaining top candidates.

For each anchor $x_i$, HeatGeo builds a candidate set

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \tag{10}$$

where $C_i^+$ contains high-probability diffusion neighbors, $C_i^h$ contains teacher hard negatives, and $C_i^r$ contains random negatives. The teacher diffusion target at scale $r$ is the diffusion mass restricted and renormalized over $C_i$:

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \qquad j \in C_i. \tag{11}$$

This target is a soft distribution over semantic relatedness, rather than a binary positive-negative label.

### Student Neighborhood Distribution

Given an anchor embedding $s_i$ and candidate embeddings $\{s_j : j \in C_i\}$, the student induces a distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_s)}, \qquad j \in C_i, \tag{12}$$

where $\tau_s$ is a student temperature. The main diffusion distillation loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}_e} \omega_r D_{\text{KL}} \left( p_{i,r}^T \| p_i^S \right), \tag{13}$$

where $B$ is a mini-batch, $\omega_r$ is the scale weight, and $\mathcal{R}_e$ is the set of active scales at epoch $e$.

### Spectral Manifold Anchoring

Diffusion matching transfers local and multi-hop neighborhoods, but a student can still fit local candidates while distorting broader community structure. HeatGeo therefore computes spectral coordinates from the teacher graph. Let the symmetric normalized graph Laplacian be

$$L^T = I - (D^T)^{-1/2} W^T (D^T)^{-1/2}. \tag{14}$$

Let $U^T \in \mathbb{R}^{N \times m}$ contain the first $m$ non-trivial eigenvectors of $L^T$, and let $u_i^T$ be the row corresponding to $x_i$. A learnable projection maps student embeddings to this spectral space:

$$z_i^S = W_{\text{spec}} s_i. \tag{15}$$

The spectral objective is

$$\mathcal{L}_{\text{spec}} = \frac{1}{|B|} \sum_{i \in B} \left\| z_i^S - u_i^T \right\|_2^2. \tag{16}$$

This term encourages the student to retain global teacher topology without requiring hidden states or teacher forward passes during student training.

### Training Objective and Curriculum

HeatGeo includes a lightweight pointwise anchor for stability:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|} \sum_{i \in B} \left(1 - \cos(W_a s_i, t_i)\right), \tag{17}$$

where $W_a$ maps student embeddings to the teacher dimension. When paired views are available, we also use a contrastive task loss $\mathcal{L}_{\text{task}}$ implemented as InfoNCE over two student encodings of each example pair.

**Algorithm 1** HeatGeo Training

**Require:** Corpus $\mathcal{D}$, teacher $T$, student $S_\theta$, graph size $k$, diffusion scales $\mathcal{R}$
Cache teacher embeddings $t_i = T(x_i)$ for all $x_i \in \mathcal{D}$
Build a mutual $k$NN teacher graph $W^T$
Compute transition matrix $P^T = (D^T)^{-1} W^T$
Precompute sparse diffusion candidates and targets $p_{i,r}^T$
Compute spectral coordinates $u_i^T$ from the graph Laplacian
**for** each training epoch **do**
  Activate diffusion scales according to the curriculum
  **for** each mini-batch $B$ **do**
    Encode anchors and candidate texts with $S_\theta$
    Compute student candidate distributions $p_i^S$
    Compute $\mathcal{L}_{\text{diff}}$, $\mathcal{L}_{\text{spec}}$, $\mathcal{L}_{\text{anchor}}$, and $\mathcal{L}_{\text{task}}$
    Update $\theta$ by minimizing $\mathcal{L}_{\text{HeatGeo}}$
  **end for**
**end for**
**return** Distilled student encoder $S_\theta$

The final objective is

$$\begin{aligned} \mathcal{L}_{\text{HeatGeo}} = {} & \lambda_{\text{diff}} \mathcal{L}_{\text{diff}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} \\ & + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_{\text{task}} \mathcal{L}_{\text{task}}. \end{aligned} \tag{18}$$

We use a diffusion-scale curriculum: the first epoch uses one-step diffusion, the second epoch adds two-step diffusion, and later epochs add the full scale set and spectral anchoring. This avoids forcing long-range, smoother structure before the student has learned reliable local neighborhoods.

## Experiments

### Experimental Setup

**Models.** Our implementation uses a compact BERT/MiniLM-style student encoder and an LLM-based teacher embedding model. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher model is removed from GPU memory; HeatGeo training uses cached teacher embeddings, precomputed diffusion targets, and spectral coordinates.

**Training Data.** We train on text-pair data assembled from multiple semantic tasks. Each training row provides either a text pair or a single text field depending on the task format. HeatGeo constructs the teacher graph over the training examples, using the anchor text for candidate neighborhoods.

**Evaluation Tasks and Metrics.** We evaluate the distilled student on three task families. For semantic textual similarity, we report Spearman correlation on SICK, STS12, and STS-B. For text classification, we freeze the encoder, train a logistic regression classifier on top of embeddings, and report accuracy and macro-F1 on Banking77, Emotion, and Tweet. For pair classification, we threshold cosine similarity and report accuracy, macro-F1, precision, recall, and average precision on MRPC, SciTail, and WiC.

**Baselines.** We compare HeatGeo with the undistilled student, pointwise teacher anchoring, mini-batch relational distillation, cross-tokenizer alignment methods such as CDM and DSKD, EMO-style token/attention alignment, Stella-style multi-faceted embedding distillation, TALAS teacher-anchored layer alignment, and the full teacher embedding model when feasible.

**Implementation Details.** Unless otherwise specified, HeatGeo uses graph size $k = 50$, graph temperature $\tau_g = 0.1$, diffusion scales $\{1, 2, 4\}$, candidate size $64$, $32$ diffusion positives, 16 hard negatives, 16 random negatives, spectral dimension 16, and five training epochs. The scale weights are $(1.0, 0.5, 0.25)$, giving higher weight to sharper local diffusion. The loss weights are $\lambda_{\mathrm{diff}} = 1.0$, $\lambda_{\mathrm{spec}} = 0.1$, $\lambda_{\mathrm{anchor}} = 0.05$, and $\lambda_{\mathrm{task}} = 0.001$.

## Main Results

Table 1 provides the planned reporting format. The key comparison is whether HeatGeo improves downstream embedding quality relative to local distillation baselines while retaining the efficiency of the compact student. We expect gains to be most visible on tasks that depend strongly on neighborhood geometry, such as semantic textual similarity and pair classification.

## Ablation Study

We isolate the contributions of HeatGeo with the ablations in Table 2. Removing diffusion reduces the method to task loss plus weak anchoring. Removing spectral anchoring tests whether local diffusion alone preserves global topology. Removing the curriculum tests whether long-range diffusion introduced from the start over-smooths early training. Candidate-set ablations test the role of hard and random negatives.

# Discussion

HeatGeo is designed around the observation that embedding distillation should match the object that retrieval systems actually use: corpus-level neighborhood geometry. The method has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that HeatGeo requires a corpus-level preprocessing step. Building a teacher graph is worthwhile when the same training corpus supports substantial distillation, but it adds overhead compared with purely mini-batch objectives. HeatGeo also depends on the quality and coverage of the teacher graph: if the teacher embeddings contain spurious neighborhoods, diffusion may propagate those errors. Future work can study approximate nearest-neighbor construction, adaptive diffusion scales, and online graph refresh for changing corpora.

# Conclusion

We introduced HeatGeo, a corpus-level heat-kernel geometry distillation framework for compact text embedding models. Instead of matching only teacher vectors, token states, or mini-batch similarity snapshots, HeatGeo distills multi-scale semantic diffusion over a teacher-induced mutual $k$NN graph and anchors global topology with spectral coordinates. This directly targets the neighborhood geometry that embedding models use in retrieval, clustering, and transfer settings while requiring only cached teacher embeddings and no student architecture changes. The resulting paper framework positions HeatGeo as a practical and retrieval-aligned alternative to local embedding distillation objectives.

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Table 1: Main evaluation results across teacher–student pairs. Classification tasks report F1, pair classification tasks report average precision (AP), and STS tasks report Spearman correlation. Avg-In averages in-domain tasks, Avg-Out averages out-of-domain tasks, and Avg. reports the overall average.

| | **Classification (F1)** | | | **Pair Classification (AP)** | | | **STS (Spearman)** | | | **Domain Avg** | | **Avg.** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Banking77 | Tweet | Emotion* | MRPC | SciTail | WiC* | SICK | STS12 | STSB* | Avg-In | Avg-Out | Avg. |
| **BGE-M3 → BERT-base** | | | | | | | | | | | | |
| Teacher | – | – | – | – | – | – | – | – | – | – | – | – |
| Student base | – | – | – | – | – | – | – | – | – | – | – | – |
| CDM | – | – | – | – | – | – | – | – | – | – | – | – |
| DSKD | – | – | – | – | – | – | – | – | – | – | – | – |
| Jasper and Stella | – | – | – | – | – | – | – | – | – | – | – | – |
| DistillCSE | – | – | – | – | – | – | – | – | – | – | – | – |
| EMO | – | – | – | – | – | – | – | – | – | – | – | – |
| TALAS | – | – | – | – | – | – | – | – | – | – | – | – |
| **HeatGeo** | – | – | – | – | – | – | – | – | – | – | – | – |
| **Qwen-Embedding-0.6B → BERT-base** | | | | | | | | | | | | |
| Teacher | – | – | – | – | – | – | – | – | – | – | – | – |
| Student base | – | – | – | – | – | – | – | – | – | – | – | – |
| CDM | – | – | – | – | – | – | – | – | – | – | – | – |
| DSKD | – | – | – | – | – | – | – | – | – | – | – | – |
| Jasper and Stella | – | – | – | – | – | – | – | – | – | – | – | – |
| DistillCSE | – | – | – | – | – | – | – | – | – | – | – | – |
| EMO | – | – | – | – | – | – | – | – | – | – | – | – |
| TALAS | – | – | – | – | – | – | – | – | – | – | – | – |
| **HeatGeo** | – | – | – | – | – | – | – | – | – | – | – | – |
| **Qwen-Embedding-4B → BERT-base** | | | | | | | | | | | | |
| Teacher | – | – | – | – | – | – | – | – | – | – | – | – |
| Student base | – | – | – | – | – | – | – | – | – | – | – | – |
| CDM | – | – | – | – | – | – | – | – | – | – | – | – |
| DSKD | – | – | – | – | – | – | – | – | – | – | – | – |
| Jasper and Stella | – | – | – | – | – | – | – | – | – | – | – | – |
| DistillCSE | – | – | – | – | – | – | – | – | – | – | – | – |
| EMO | – | – | – | – | – | – | – | – | – | – | – | – |
| TALAS | – | – | – | – | – | – | – | – | – | – | – | – |
| **HeatGeo** | – | – | – | – | – | – | – | – | – | – | – | – |

Table 2: Ablation template for HeatGeo. Report average score across task groups or a selected validation suite.

| Variant | Average Score |
|---|---|
| HeatGeo full | – |
| w/o diffusion loss | – |
| w/o spectral anchoring | – |
| w/o pointwise anchor | – |
| w/o diffusion curriculum | – |
| w/o hard negatives | – |
| w/o random negatives | – |
| single scale $r = 1$ | – |
| single scale $r = 4$ | – |


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